

Xiaochuan Li

M.Sc. Human and AI · University of Technology Nuremberg

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EDUCATION

M.Sc., Human and AI

2025 to 2027

University of Technology Nuremberg (UTN)

Research focus. Whether language models build the semantic structures formal linguistics proposes (modal force, morphological composition, genericity), or only their surface. I turn each theory's claim into a prediction that controlled intervention and interpretability can decide, testing what competing theories predict rather than crowning one correct.

Coursework. Cognitive psychology · interpretability · deep learning · epistemology.

German language program

2024 to 2025

University of Marburg · intensive German for graduate study

B.A., Philosophy

2020 to 2024

Shenzhen University · College of Humanities

Senior thesis: *On Leibniz's Logical Theory* (advisor: Zang Yong)

Coursework: logic, history of philosophy, analytic philosophy, mathematical logic; electives in cognitive psychology and linguistics. Shapes my focus on reasons, causality, and what licenses an inference.

RESEARCH: THREE CASE STUDIES

One question, three kinds of word where natural language's logic looks unlike the textbook's. Each turns a formal-semantic hypothesis into a causal test read with interpretability.

MODUS: modality

Master's thesis · pilot run

Is modal force a set of possible worlds (Kratzer) or a degree of certainty (Lassiter)? A pilot builds a certainty direction from evidence alone, with no modal word; ablating it on a *might* sentence pushes its certainty toward *must* (+0.15), while a base-rate control does not. GRADUS, a graded epistemic-force dataset, turns the probe into a calibration diagnostic.

[Project page](#) ↗

HanGL: morphological composition

personal · pilot done

Is a Sino-Korean word's meaning composed from its Hanja morphemes or memorized whole? Hangul hides the morphemes; Hanja is the answer key the model never sees. A wrong-Hanja intervention shows latent grounding scales with model capability; the forward step tests morphological decomposition against the human psycholinguistics of Sino-Korean.

[Project page](#) ↗ · [GitHub](#) ↗

ORDO: genericity

in design

"Birds fly" is not "all birds fly": a penguin does not falsify it. That default-with-exceptions is an inference classical logic cannot state. Do generics and modals share one ordering-source mechanism for defaults, or two? A layered novelty-of-exception design where either outcome is a real finding.

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COLLABORATIONS

Latent Control States · DFKI

Does prompt framing shift a causally relevant latent state, or only the output? My contribution: I designed and built the controlled two-agent data pipeline the probing study runs on.

Cross-lingual ProtoBias · an established NLP group

Does a vision-language model track meaning, or culture-specific prototype shortcuts? My contribution: the cross-lingual evaluation across seven languages that extends the group's forthcoming paper.

[GitHub](#) ↗

SKILLS

Both the engineering my research needs (controlled linguistic data generation, transformer model evaluation, intervention-based causal testing, reproducible pipelines) and the foundations it draws on: cognitive psychology, linguistics, and logic.

Programming & data

Python (NumPy, pandas, scikit-learn), Git, LaTeX.

ML / NLP tooling

PyTorch and HuggingFace Transformers (working proficiency): model inference, dataset pipelines, LoRA fine-tuning.

Causal & interpretability methods

Controlled do-interventions, causal graphs, and shortcut-severing audits; developing behavioral path-tracing.

Cognitive psychology

Working foundations in concept learning, cognitive development, and human reasoning, including the developmental literature on modal cognition (Leahy & Carey), which grounds the emergence question the thesis asks.

Linguistics & logic

Formal semantics (possible-worlds, modal quantification), modal and predicate logic, and morphology/syntax across isolating, fusional, and agglutinative types; grounded in a philosophy background in logic and philosophy of language.

Linguistic evaluation

Designing controlled stimuli for morphology, agreement, dependency structure, and word order across Chinese, German, English, and Japanese.

LANGUAGES

Mandarin Chinese (native) · German (C1) · English (C1), spanning isolating, fusional, and analytic types. Learning Japanese, an agglutinative head-final system, to extend the typological span my linguistic work draws on.